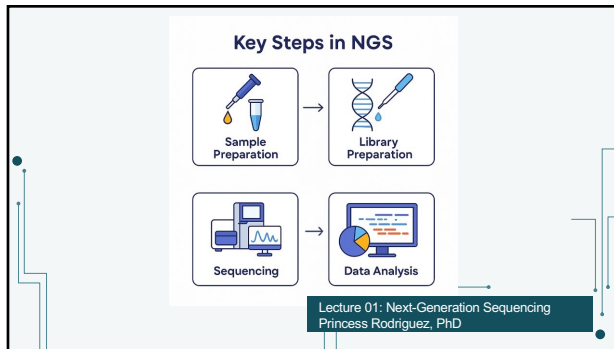
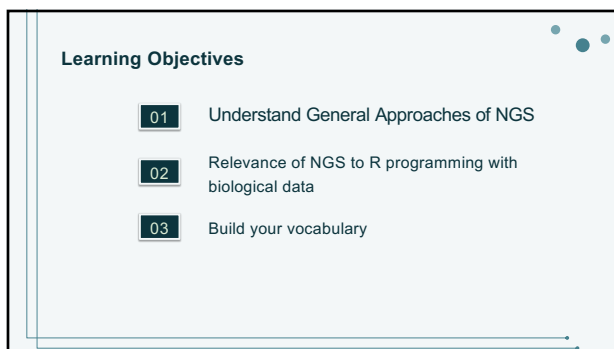
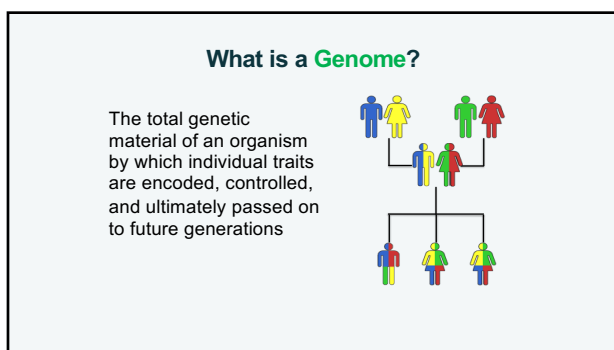




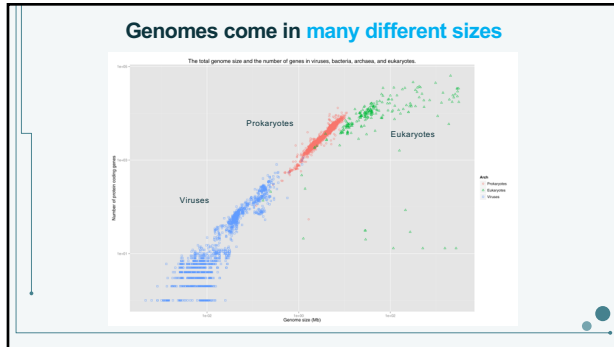
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2

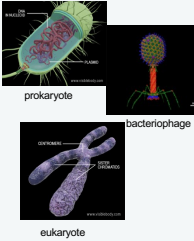


3



4

Genomes comes in different shapes



Primarily DNA, but can be RNA in the case of some viruses

Some genomes are circular, others linear

Can be organized into discrete units (chromosomes) or freestanding molecules (plasmids)

Labels in image: prokaryote, bacteriophage, eukaryote

5

Challenges of DNA

DNA has only four letters

They are strung together with NO obvious punctuation

There are no signals to say "a gene starts here"

6

```

AGGTTAGATTATGCCCGAGGGCGCCCGAGCCGAAATTTTAAATGCAGGTTTAAATAGTTTAGAGC
CTGTGGGCTTCCATGGCTTGGTTCTGCTGTTCTTCACTGGGGAATGGGGGACCTGGGAGCTTG
TGATGGGGCTGTCTCCACCTCTGTAAATCCAGGAGTCAAGATGACAAATCTGTCAATTCGGGGG
ACACACTCCCCGAGGAAGGGCTTGCAGGAGGGCAGAGAGCTTCTGTGGGATGGCAGGGAGT
GGAGAAGGGCAGGGGGGCGCAGAGCAGGAGCAGCTTCTGCTCTGGGTGGGACAGTGATCCCCA
CTGGGGAATGGCAAAGCCCCATGCTCTCTGTTACCTGGATGGGTGGCACTGGGGGAGGCAAT
GGGGCTGCAGGAGCCCTGTGTGTCAGCCCTCCCTGCCAGCATCCATCTCCAGGAGGGCCCC
CAGGGCAGGTAAGTGCCAGGTCCCCCTCAGCTCACGTTGTCTTCCCTTGACGAGCCCTCC
CACTCCCGAACCAGTGCATGCTGATGAGTAGATGACGCCGGGACTCCTCGGCTGGAAGGC
CTTGTTCACCTGGGAGGGGGGAGGAGCGGAGGAGAGAGGGGAGGTCACTCTATTATTTACCT
TCAGCAAGGATTTCCCAAATGCCCGGCCAGTCTTCAACCAAGATCTTACAAAACACAG
ATACTCAAGGTGAAAATCAAAACCCAGACAGGCTCAAAACAAATGAACACACCATCACTCAG
AGAGGACGGGTGAACATTTTGGGGTCTCTTCAATAACTGTTGACCCAACTGAGACACACAGG
GAGATTCTACTTTTGAAGGAATCTCACTCTGTCAACCGAGGTGGAGTGCAGTGGCGCATCT
CGGCTTACTGCAACCTCTGCTCCAGGTTCAAGCGATTCTCCACCTCAGCCTTTGAGTAGCT
GGGATTACAGGTGTGTGCCACCACTGGCTACTTTTGTGTTTTAGTAGAGACGGGGTTTCGCC
ATGTTGGCAGGGTGGTTGAACTCTGAGCTCAGGTGATCCGCCACCTCGACCTCTCAAAAG
TCTGGGATAACAGGCATGAACCACTGCGCCGGGCTGGGAGATCTAATTTCTCCGGTGAAT
AGAATGTGCTATCTGCTCAGAGAGGCACTCTCTTCTGACAGGAGCATTTCTTTTCAGAT
GGGGGTGTCTCACTCTGTGCCAGGGGAGTGAGCGGCAATCATGGCTCACTGCGACCT
TGACCTCTGGGCTCAGTGATCTCCCACTCAGCTCTCTGAGTAGGTTGGACACAGGTGCAT
ACCACTAGGCCAGCCCTGACAGTCTCTTTTCTGTTGTTCTGAGACAGGCTCACTCTATT
GCCCAGGCTGGGTGAGTGGCATGATACGGCTCACTGACGCTCAACCTCCAGGCTTAGGTG
ATCTCCCACTCACTCAGGCTCCAGGTAGCGGGGACTACAGGTACACATCACTGCTGGCT
AATTTTGTATTGTTGTAGAGATGGGGTTTCGCCATGTTGGCAAGTGGTCTGAACCTCTGG
    
```

7

**Challenges
of DNA**

DNA has only four letters

**Challenges
of DNA**

They are strung together with
NO obvious punctuation

**Challenges
of DNA**

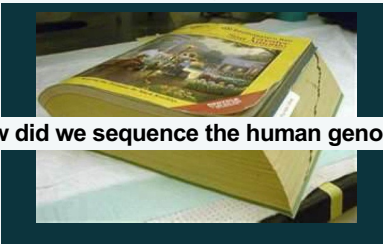
There are no signals to say
"a gene starts here"

**Challenges
of DNA**

Its large size

8

How large?

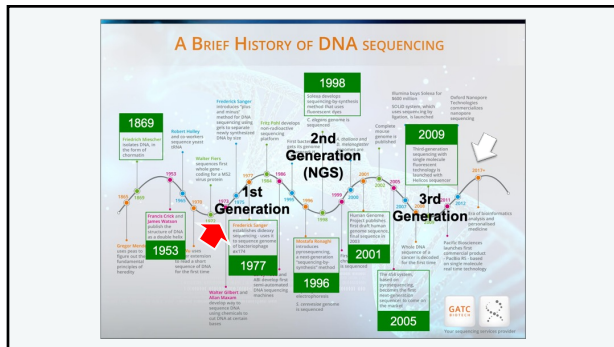


How did we sequence the human genome?

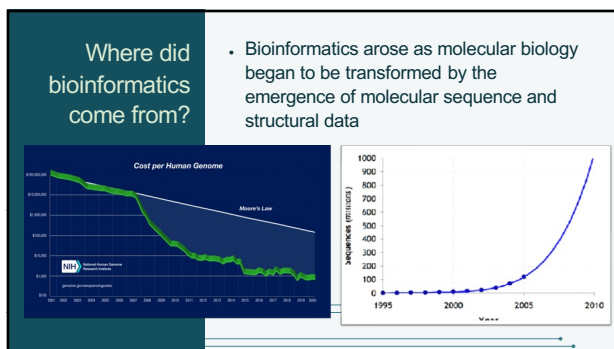


If compiled in books, it would fill an estimated 200 volumes the size of a Manhattan telephone book (at ~1000 pages each) and reading it would require 26 years working around the clock.

9



10



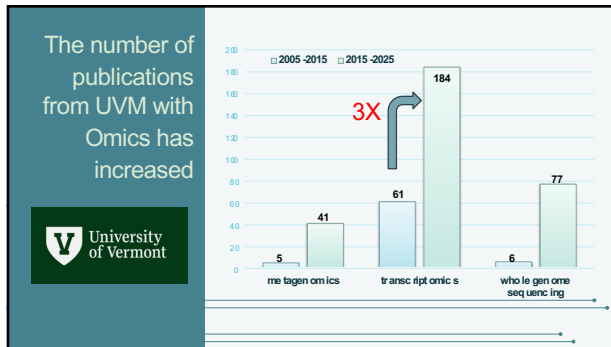
11

What is Bioinformatics?

The use of biological databases and computational programs to analyze proteins, genes, and the complete collection of DNA that comprises an organism (the genome)

... Bioinformatics is a hybrid of biology and computer science
... Bioinformatics is computer-aided biology!

12



13

How do we deal with it?

We accept that the needs are changing

14

	1980s:	1990s – 2000s:	2010 and beyond....
Molecular	• RFLP • PCR	• microarrays • RNAi	• RNA-Seq • Single-cell • Spatial
Computational	• Word processing	• BLAST • Databases	• R programming • Linux/UNIX • Python

15

In the lab, someone does this....

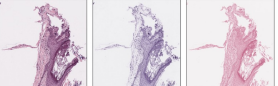


... this might be the next step

Stain Deconvolution

```

101: def stain_deconvolution(image, targets, targets_colors, targets_names):
102:     """
103:     Stain deconvolution using the Macenko method.
104:     """
105:     # Convert image to Lab color space
106:     lab_image = cv2.cvtColor(image, cv2.COLOR_BGR2LAB)
107:     # Convert targets to Lab color space
108:     lab_targets = []
109:     for target in targets:
110:         target_color = cv2.cvtColor(target_color, cv2.COLOR_BGR2LAB)
111:         lab_targets.append(target_color)
112:     # Convert targets to Lab color space
113:     lab_targets_names = []
114:     for target_name in targets_names:
115:         lab_targets_names.append(target_name)
116:     # Perform stain deconvolution
117:     lab_image = macenko_stain_deconvolution(lab_image, lab_targets, lab_targets_names)
118:     # Convert image back to RGB color space
119:     rgb_image = cv2.cvtColor(lab_image, cv2.COLOR_LAB2RGB)
120:     return rgb_image
    
```



macenko method



16



Next Generation Sequencing Overview

17

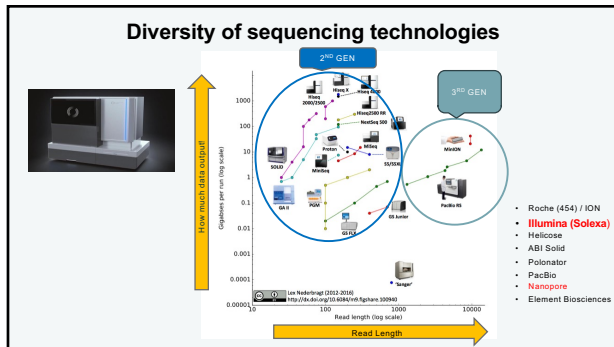
What is NGS?

- Is a high-throughput DNA sequencing methodology that makes use of parallelization to process up to half a million sequences concurrently
- Often taken to mean 2G technologies

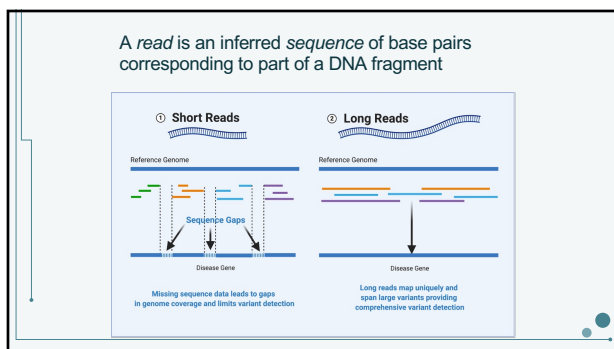
First generation sequencing (chain termination)	Second generation sequencing (massively parallel)	Third generation sequencing (single molecule)
Sanger	Illumina	Nanopore
DNA purification	DNA purification	DNA purification
Fragmentation	Fragmentation	Fragmentation
Cloning	Adaptor ligation	Adaptor ligation
Amplification	Amplification & Sequencing	Sequencing
Sequencing	Sequencing	Sequencing
DNA template	Flow cell	Nanopore flow cell
Primer	Bridge amplification	Sequencing
DNA template	Flow cell	Sequencing
Detection	Detection	Detection
ACGT	ACGT	ACGT
- Sequencing by synthesis - Read length: 100-1000 nucleotides - Accuracy: 99.9% - Real-time analysis: No - Output data: low output	- Sequencing by synthesis - Read length: 35-600 nucleotides - Accuracy: 99.9% - Real-time analysis: No - Output data: high output	- Single molecule sequencing - Read length: 100-2 million nucleotides - Accuracy: ~90-95% - Real-time analysis: Yes - Output data: high output

Next Generation Sequencing (NGS)

18



19



20

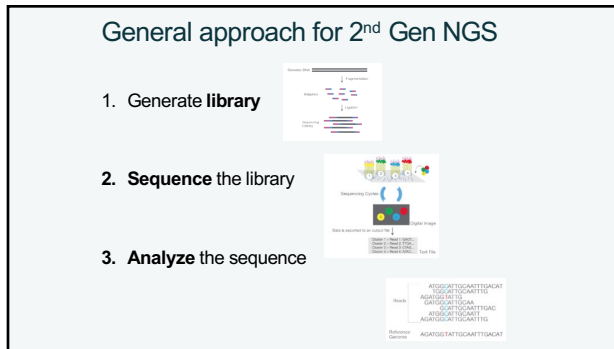
Next-generation sequencing (= 2nd generation)

- unifying characteristics of the different NGS platforms:
- short reads (50-250 bp) and short fragments (250-1000 bp)**
 - Includes HiSeq, NextSeq, NovaSeq platforms
 - Run Time varies but can be up to > 48hrs
 - Maximum Output is ~25 to 400 million reads per lane

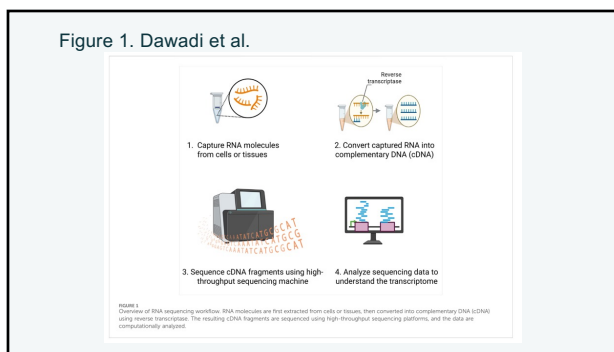




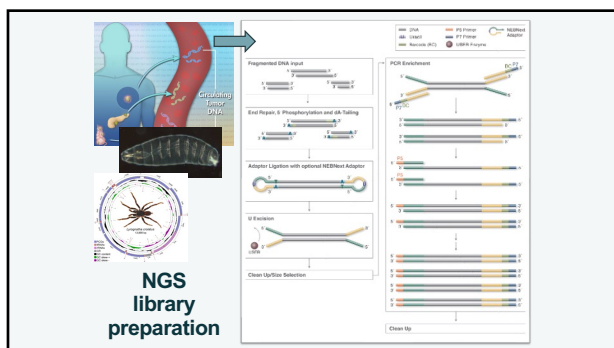
21



22



23



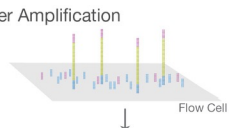
24

Libraries are then loaded onto a flow cell

- The flow cell is a glass slide with lanes
- Libraries are loaded into a flow cell and the fragments are hybridized to the flow cell surface

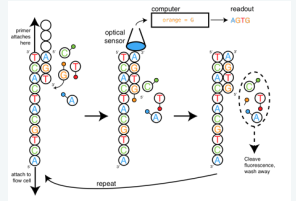


B. Cluster Amplification



Flow Cell

25



Then sequencing begins!

- Have fluorescent-labeled nucleotides
- Each dNTP has a corresponding color
- During each **cycle**, a labeled dNTP is added to the growing chain
- An image is then taken
- Then the fluorescent dye is cleaved to allow for the next nucleotide to be incorporated in the next **cycle**

26

Problem

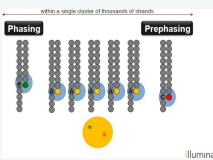
- Sequencing by synthesis progressively becomes out-of-sync therefore, signal quality deteriorates as the read-length grows
- At each cycle, accumulates noise

Solution

- Sequence only (up to 300bp)

The human genome is 3 billion base pairs i.e. this is a lot of reads generated !!

within a single cluster of thousands of reads



illumina

27

Third Generation Sequencing

- Typical characteristics:
 - No amplification step
 - **Long (1,000bp+) sequence reads**
 - Real-time monitoring of nucleotide incorporation
 - Reads the nucleotide sequence at the **single molecule** level
 - Often associated with nanopore technology



28

Oxford Nanopore Technology (ONT) sequencing

- Sequencers can be pocket-sized, USB connectable, cost \$150 to \$1,000
- Generates incredibly long reads: read length are typically between 10,000bp & as long as 1,000,000 bp!

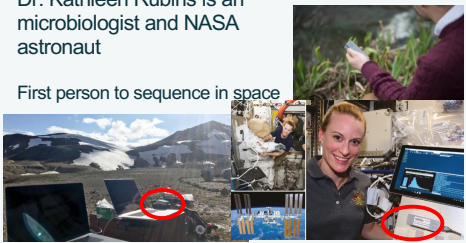


29

Dr. Kathleen Rubins is an microbiologist and NASA astronaut

First person to sequence in space

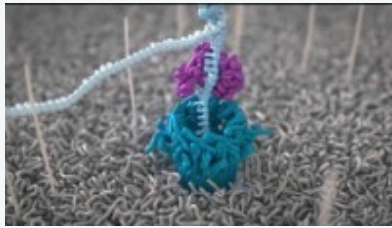
"Analysis of anything, by anyone, anywhere"



<https://www.nasa.gov/astronauts/biographies/kathleen-rubins/biography>

30

- A strand of DNA is passed through a nanopore
- The magnitude of current is changed as the bases G, A, T and C pass through the pore in different combinations



31

The FASTQ format.



A “standard” format for storing and defining sequences from next-generation sequencing technologies.

```
Sequence ID @SEQ_ID
Sequence GATTGGGGTTCAAGCAGTATCGATCAAATAGTAAATCCATTGTTCAACTCACAGTTT
(separator) +
Quality scores !''*(((('*+)})))})))}.1***-+''')**55CCF>>>>>CCCCCCC65
```

http://en.wikipedia.org/wiki/FASTQ_format

32

Figure 1: Workflow of the H3K4me3-based pipeline for identifying enhancers and enhancer-associated genes.

The pipeline consists of the following steps:

- Quality Control and Preprocessing:** Raw reads are processed using FastQC and Trimmomatic.
- Read Alignment and Indexing:** Reads are aligned to the reference genome using BWA-MEM. The resulting BAM file is indexed using samtools.
- Peak Alignment Processing:** Peaks are called using MACS2. The resulting peaks are filtered and annotated using bedtools and Ensembl.
- Peak Distribution:** Peaks are distributed across the genome using bedtools.
- Enrichment Analysis:** Peaks are enriched for specific genomic features using bedtools and Ensembl.
- Enrichment and Normalization:** Peaks are normalized using bedtools and Ensembl.
- Functional Annotation:** Peaks are annotated with functional information using bedtools and Ensembl.
- H3K4me3 Peak Calling:** H3K4me3 peaks are called using MACS2.
- H3K4me3 Peak Calling:** H3K4me3 peaks are called using MACS2.
- H3K4me3 Peak Calling:** H3K4me3 peaks are called using MACS2.

The pipeline identifies enhancers and enhancer-associated genes using H3K4me3 ChIP-seq data. Key steps include peak calling, peak distribution, enrichment analysis, and functional annotation. The final output is a list of enhancers and enhancer-associated genes.

33

[illegible]

The diagram illustrates the integration of Human and Microbiome data through various technologies and data analysis methods to understand their functional effects and network interactions.

Human Data Path:

- Technologies:** GENOMICS (WGS, WES)
- Data analysis:**
 - Point mutations
 - Small indels
 - Copy number variation
 - Structural variation
 - Lineage identification
- Applications:**
 - Functional effects of mutations
 - Network & pathway analysis
 - Integrative analysis

Microbiome Data Path:

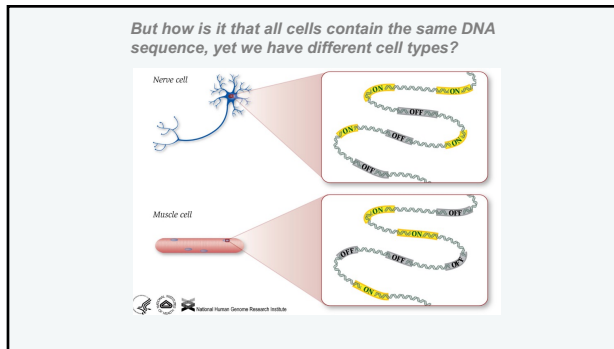
- Technologies:** TRANSCRIPTOMICS (RNA-seq), EPIGENOMICS (Methyl-seq, ChIP-Seq, ATAC-seq), METAGENOMICS (16S rRNA, ITS, Shotgun, Targeted panels)
- Data analysis:**
 - Differential expression
 - Gene fusion
 - Alternative splicing
 - RNA editing
 - Methylation
 - Histone modifications
 - Transcription factor binding
 - Bacterial profiling
 - Fungal profiling
 - Microbiome profiling
 - AMR gene profiling
 - Targeted gene analysis
- Applications:**
 - Functional effects of mutations
 - Network & pathway analysis
 - Integrative analysis

Research, Diagnostics & Healthcare:

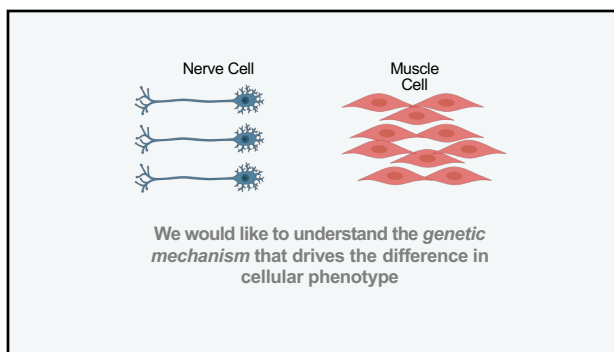
The diagram shows that both Human and Microbiome data analysis paths lead to the same set of applications: Functional effects of mutations, Network & pathway analysis, and Integrative analysis, which are ultimately used for Research, Diagnostics & Healthcare.

The diagram illustrates the hierarchy of genetic material. On the left, a blue sphere represents a **Cell**. A yellow cone expands from the cell to show a blue, X-shaped **Chromosome**. A section of the chromosome is magnified to show a green double helix structure labeled **DNA**. A bracket at the bottom indicates a specific segment of the DNA as a **Gene**.

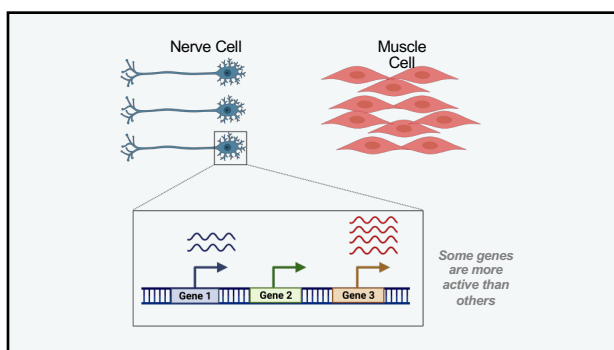
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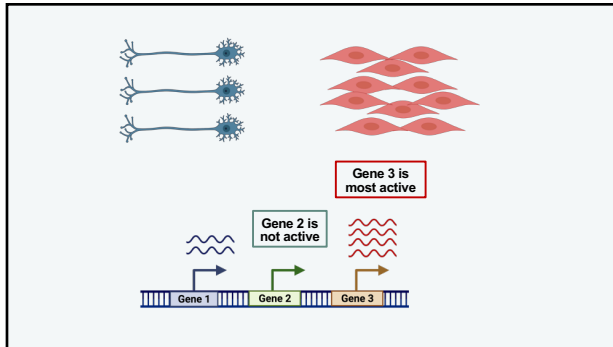
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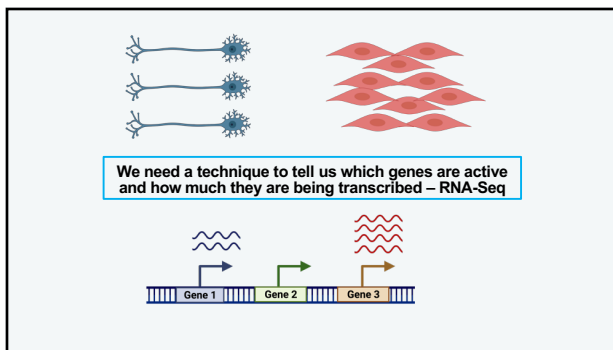
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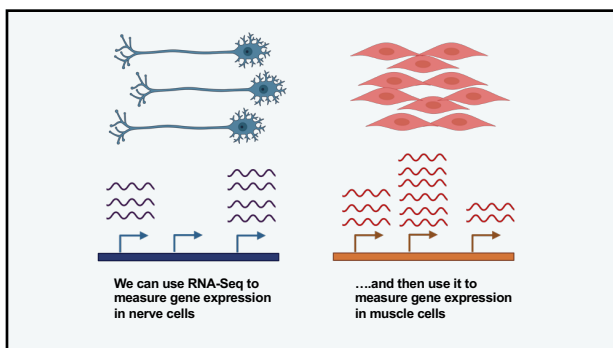
39



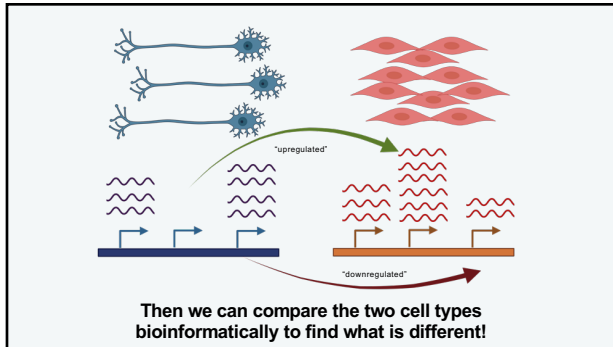
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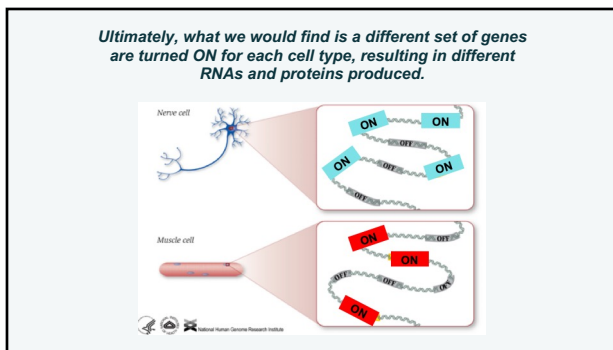
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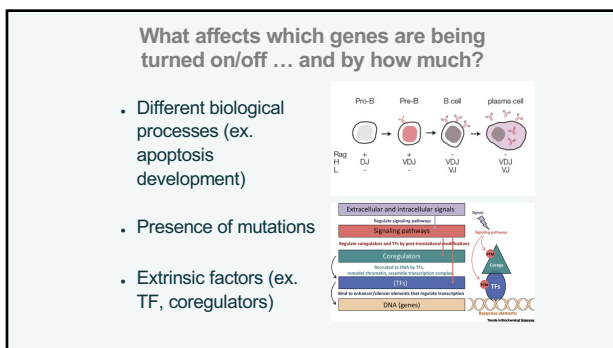
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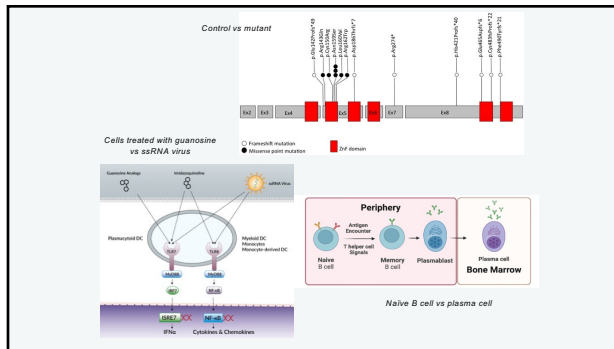
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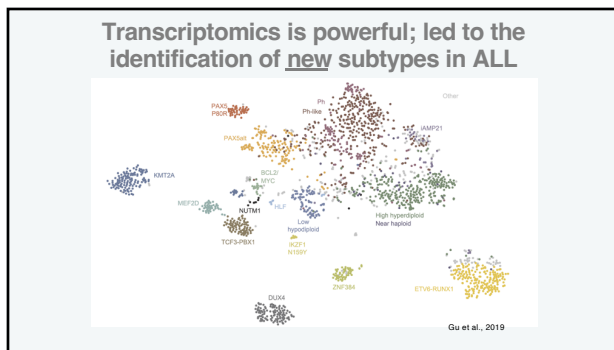
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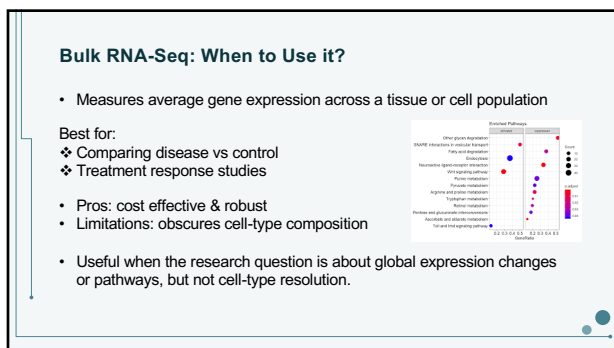
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46

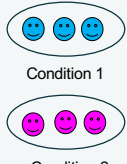


47



48

Biological Replicates



Condition 1

Condition 2

❖ To detect Differentially Expressed Genes (DEGs) between groups we should have several samples, which are also known as replicates

Differential Gene Expression: quantitative evaluation and comparison of transcript levels across treatments or groups

49

Biological Replicates

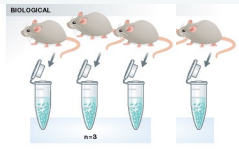
- We are not talking about technical replicates
- Assessing biological variation requires biological replicates
- Three is the standard minimum
- Yet I would recommend **four** (More is always better!)

TECHNICAL



n=1

BIOLOGICAL



n=3

50

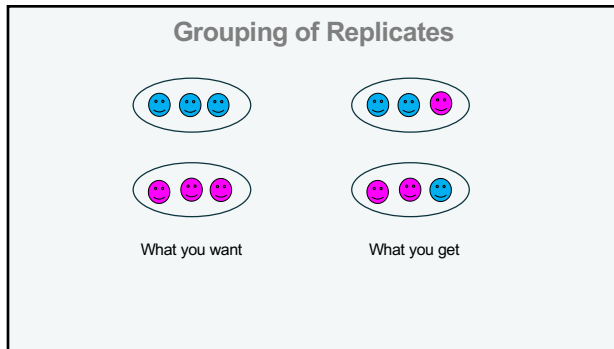
Probability of detecting DEGs

	Replicates per group		
	3	5	10
Fold change			
2	87%	98%	100%

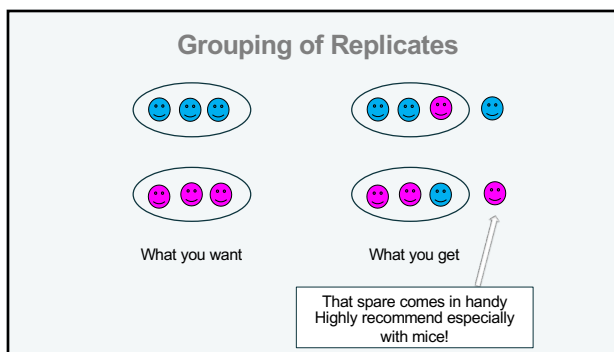
↑

PMID: 26813401

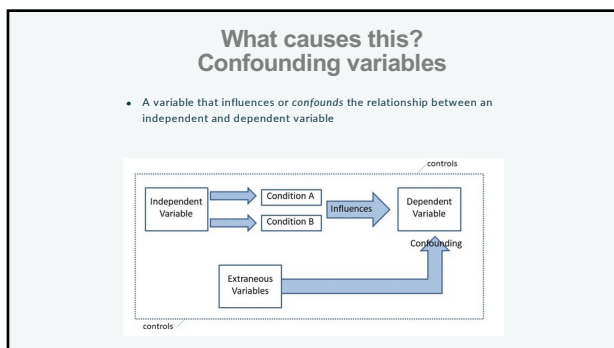
51



52



53



54

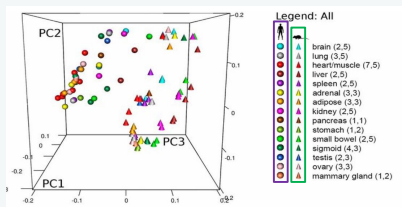
Examples of common confounding variables

- New technician is processing your samples and/or running the sequencer
- Extracting RNA with different kits
- Inappropriate multiplexing strategy
- Sequencing on multiple different types of platforms

A well-planned experiment with an additional sample, does end up saving you time and money down the road

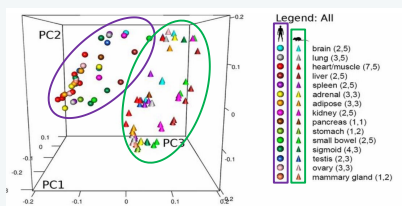
55

ENCODE reported that gene expression was likely to follow a species-specific rather tissue-specific pattern

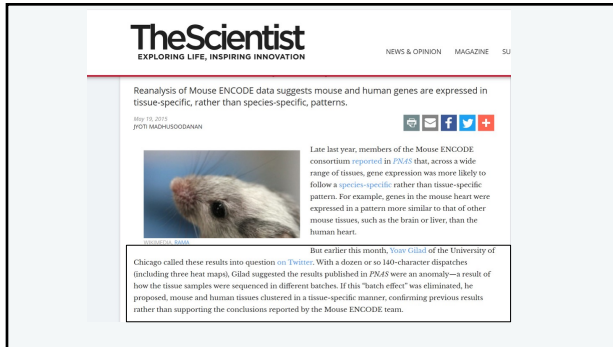


56

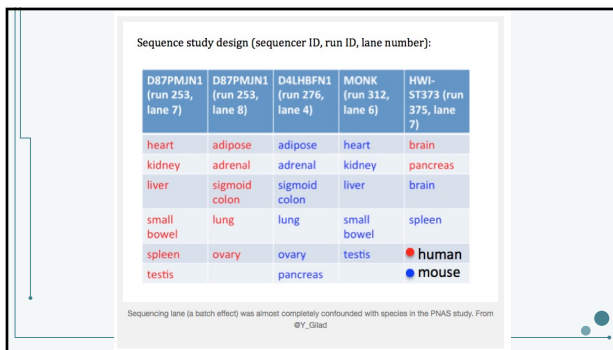
ENCODE reported that gene expression was likely to follow a species-specific rather tissue-specific pattern



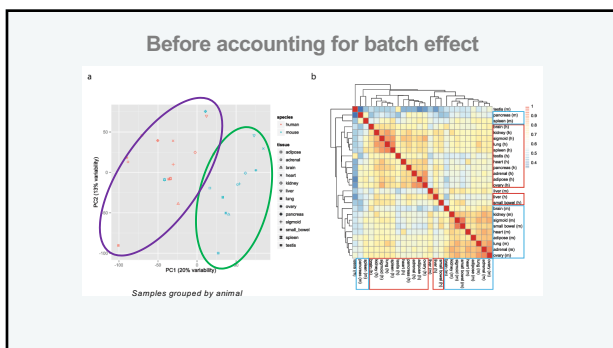
57



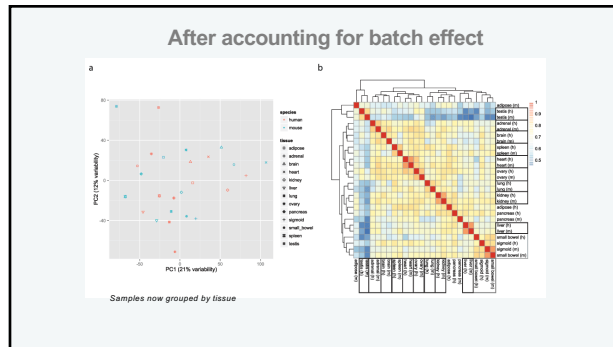
58



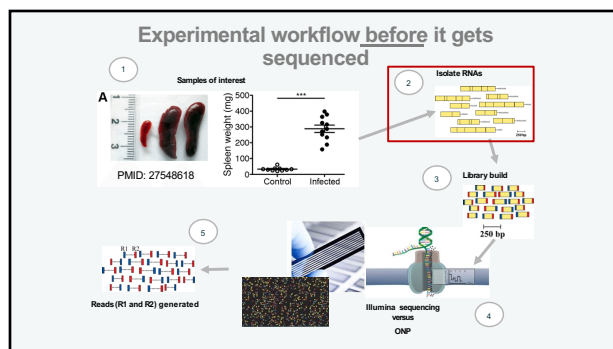
59



60



61



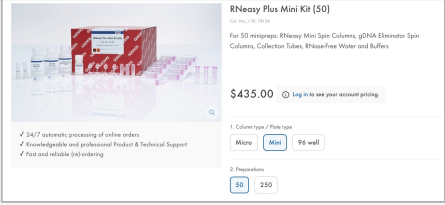
62

Working with RNA

- RNA-Seq is dependent on the isolation of pure RNA
- RNA is more labile than DNA so extra precautions should be taken

63

Isolation of RNA



RNeasy Plus Mini Kit (50)
 For 50 miniprep RNeasy Mini Spin Columns, gDNA Eliminator Spin Columns, Collection Tubes, RNase-Free Water and Buffers

\$435.00 [Log in to see your account pricing.](#)

- 24/7 automatic processing of online orders
- Knowledgeable and professional Product & Technical Support
- Fast and reliable (re)ordering

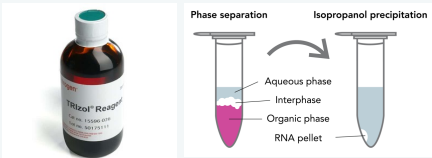
1. Column type / Plate type
 Micro Mini 96 well

2. Tissue amount
 35 250

- This kit comes everything you need for total RNA isolation including the gDNA eliminator spin columns

64

The cheaper way to isolate RNA



TRIzol® Reagent

Phase separation **Isopropanol precipitation**

Aqueous phase
 Interphase
 Organic phase
 RNA pellet

- The UVM VIGR core **does not** recommend as residual phenol will inhibit the library build and you cannot get rid of it. They will not sequence your samples if you use this mode of isolation!
- Pro-tip: maintain separate reagents and consumables.

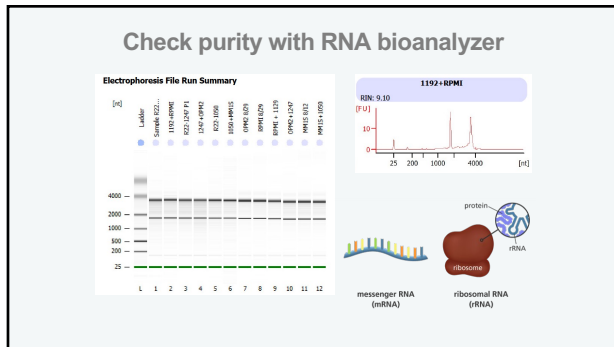
65

Other recommendations

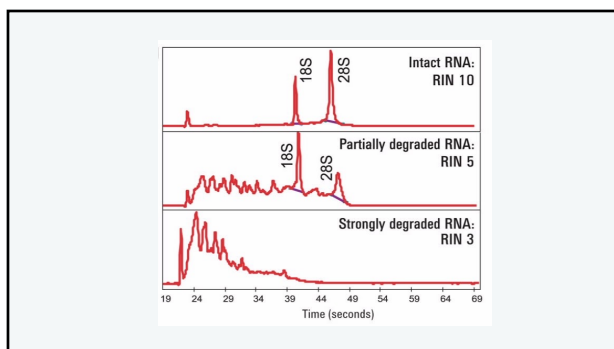
- Always keep your RNA samples on ice
- Spray bench, pipettes, centrifuge, and tip boxes with RNase Zap
- Maintain RNA integrity with **Ribolock** (ThermoFisher: EO0381)
- Once isolated, RNA is stored in the -80C



66



67

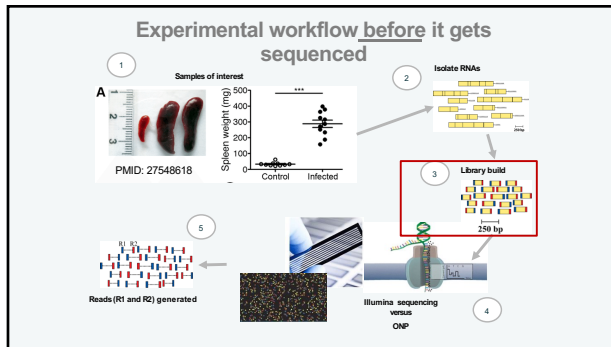


68

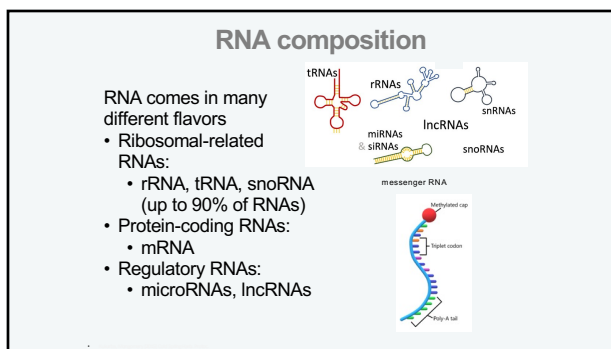
RNA Bioanalyzer submission at UVM

- Need to take a Qubit reading first not nanodrop
- Then determine whether a *pico* or *nano* chip is required based on this
- Submit 2 ul of sample to core in 0.6mL tube
- Fill out form on iLabs

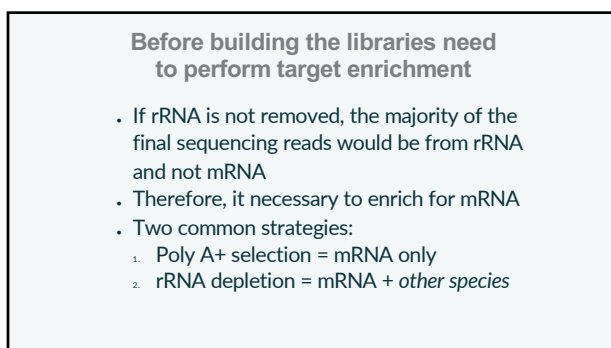
69



70



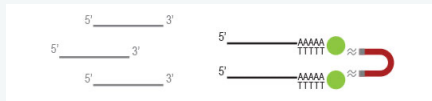
71



72

Poly-A versus rRNA depletion?

- For differential gene expression, most enrich using Poly(A)+
- However:
 - If you are aiming to obtain information about long non-coding RNA's perform ribosomal RNA depletion
 - Bacterial mRNAs are also not poly-adenylated



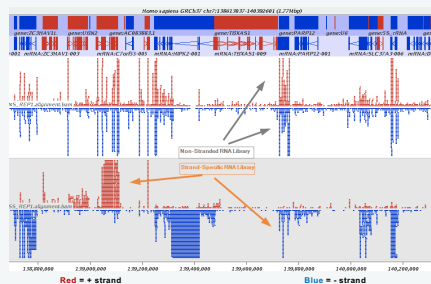
73

Strandedness

- Another consideration is whether to generate strand-preserving libraries
 - Libraries can be stranded or unstranded
- The implication of **stranded** libraries is that you could distinguish whether the reads are derived from forward or reverse-encoded transcripts

74

Strandedness



75

Motivation - Single cell level insights

Bulk RNA-seq



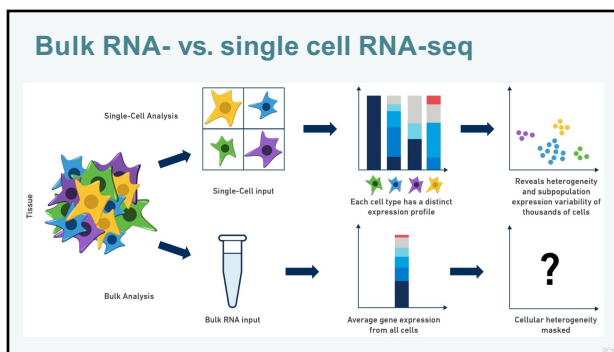
Taste the average of all fruits

Single cell RNA-seq

Taste each fruit individually



76

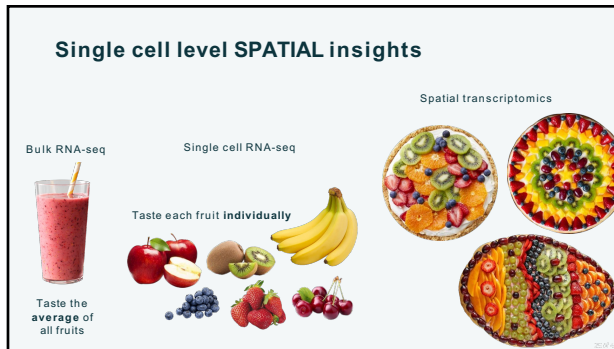


77

Single-Cell RNA-Seq

- Measures expression **cell-by-cell**
- Best for:
 - ❖ Identifying novel cell types/states
 - ❖ Uncovering tumor heterogeneity
 - ❖ Immune profiling
 - ❖ Understanding treatment-resistant populations
- Pros: High-resolution
- Limitations: Expensive, technological challenges as the data is noisy
- Select scRNA-seq when you suspect heterogeneity matters

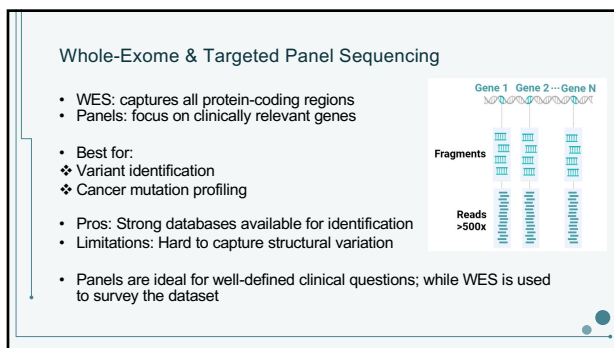
78



79

	Bulk RNA sequencing	Single-cell RNA Sequencing	High Throughput Spatial Transcriptomics
Resolution	Patient-level gene expression	Single-cell resolution	Spatial and single-cell resolution
Data Format	Aggregated gene expression from whole tissue	Gene expression at the individual level	Gene expression mapped to 2D tissue coordinates
Biological Insights	Identifies differentially expressed genes at the tissue level	Identifies cell subpopulations and heterogeneity	Reveals spatially variable genes, tissue structure, and cell-cell interactions
Advantages/Disadvantages	✓ High sensitivity for overall gene expression ✗ No spatial or single-cell resolution	✓ High resolution for individual cell types ✗ Loses spatial context and cell-cell interactions	✓ Preserves tissue architecture and spatial relationships ✗ Lower sensitivity for minimally-expressed genes

80



81

One step above this is WGS...

Whole genome sequencing	Whole exome sequencing	Targeted sequencing
- Sequencing region: whole genome - Sequencing Depth: >30X - Covers everything—can identify all kinds of variants including SNPs, INDELs and SV.	- Sequencing region: whole exome - Sequencing Depth: >50X ~ 100X - Identify all kinds of variants including SNPs, INDELs and SV in coding region. - Cost effective	- Sequencing region: specific regions (could be customized) - Sequencing Depth: >500X - Identify all kinds of variants including SNPs, INDELs in specific regions - Most Cost effective

82

16S & Metagenomic Sequencing

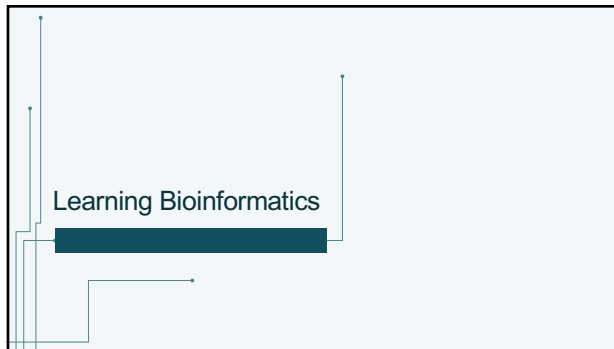
- A major reason why researchers perform sequencing is to **differentiate between multiple bacterial species within a sample**. The ability to identify novel, rare, or unexpected pathogens at a **species and strain level** is of high priority.
- 16S: bacterial composition at genus level
- Shotgun metagenomics: species level & functional pathways
- 16S is inexpensive; shotgun is comprehensive

83

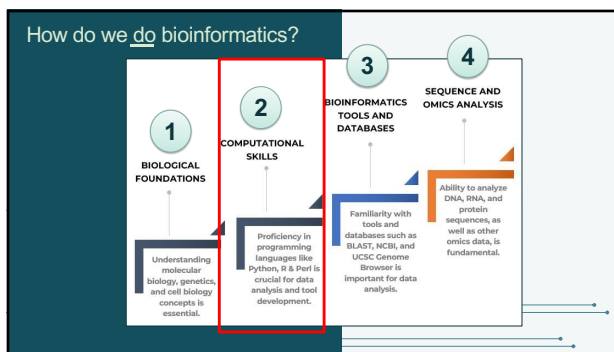
Choosing the Right Design

- Need for matching your research question with the correct modality
- "Which pathways change after treatment?" **mseq**
- "What cell type drives disease?" **scrnaseq**
- "What mutations define relapse?" **WES**
- "How does the microbiome differ between groups?" **metagenomics**

84



85

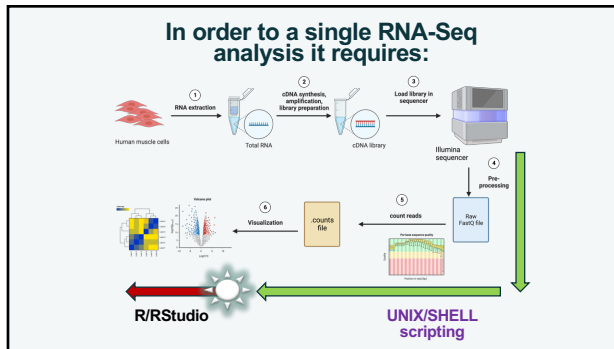


86

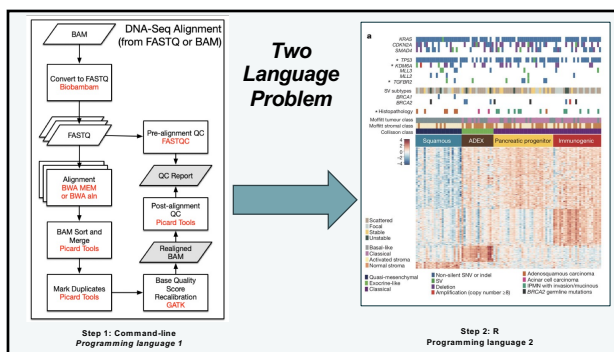
What are programming languages?

- Programming languages are tools
- They make it possible to instruct computers
- In biomedical research, the prevailing languages are R, Python, and command line

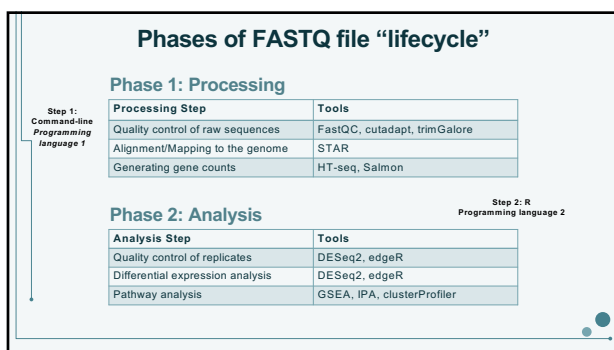
87



88

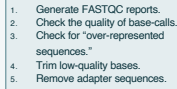


89



90

Quality Control Steps

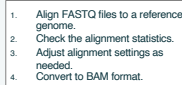


```
fastqc 2D_WT_shGFP_E2_R50_001_S10_R1_001.fastq.gz  
  
trim_galore --paired --Illumina 2D_WT_shGFP_E2_R50_001_S10_R2_001.fastq.gz  
2D_WT_shGFP_E2_R50_001_S10_R2_001.fastq.gz
```

1. Generate FASTQC reports.
2. Check the quality of base-calls.
3. Check for "over-represented sequences."
4. Trim low-quality bases.
5. Remove adapter sequences.

[illegible]

Alignment Steps



Where do you get the reference genome files? https://www.ncbi.nlm.nih.gov/assembly/GCF_000001405.26

STAR Alignment Summary:

```

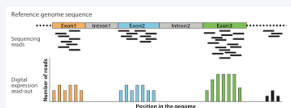
Number of input reads | 35202511
Average input read length | 299
UNIQUE READS:
Uniquely mapped reads number | 33777023
Uniquely mapped reads % | 95.95%
Average mapped length | 298.68

```

1. Align FASTQ files to a reference genome.
2. Check the alignment statistics.
3. Adjust alignment settings as needed.
4. Convert to BAM format.

[illegible]

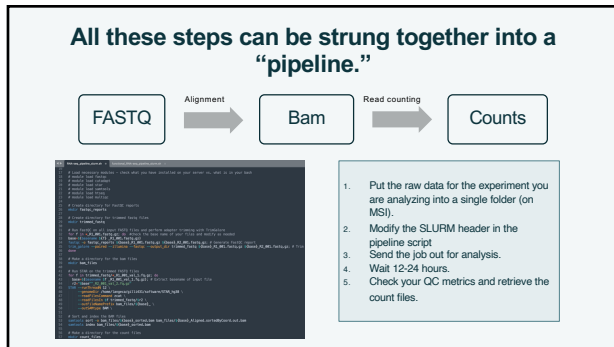
Gene Counting Steps



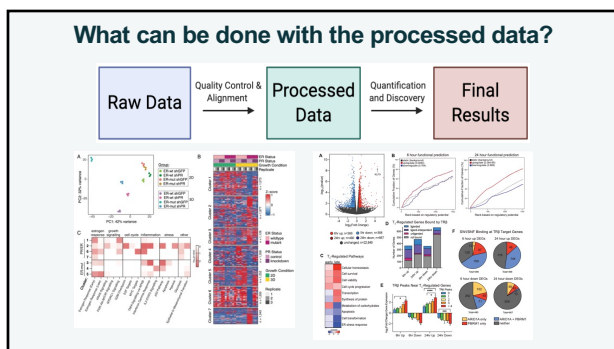
```
htseq-count -f bam -r name -s reverse -m union -i gene_id bam_files/${base}_sorted.bam
~/software/gencode.v38.primary_assembly.annotation.gtf > count_files/${base}.counts
```

1. Check that bam files are sorted and indexed
2. Run HT-seq on bam files.
3. Save .cnts files for analysis.

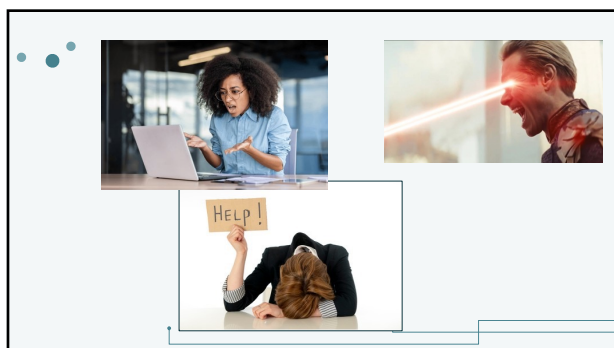
[illegible]



94



95



96
